

YUMA 27 ENE, AZ, USA

WMO: 722790

Lat: 32.8350N

Lon: 114.1884W

Elev: 189

StdP: 99.08

Time Zone: -7.00 (W07)

Period: 08-19

WBAN: 53154

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	2.2	4.1	-21.0	0.6	21.5	-18.2	0.8	19.5	7.6	14.4	6.7	14.8	0.9	N/A	0.414

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	43.6	20.2	42.4	20.2	41.3	20.0	24.1	33.5	23.6	33.9	23.0	34.3	3.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
21.7	16.8	28.0	20.6	15.6	28.5	19.5	14.5	29.9	72.9	33.2	71.0	33.3	69.0	33.9	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.0	6.0	5.3	-1.9	46.3	1.9	1.4	-3.3	47.3	-4.4	48.1	-5.5	48.8	-6.9	49.8		
(5)			-5.0	25.3	1.6	0.7	-6.1	25.8	-7.1	26.2	-7.9	26.6	-9.1	27.1		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	23.3	13.7	15.1	19.0	21.9	25.1	31.1	34.1	33.9	30.7	24.1	17.9	12.7		
	DBStd	8.11	3.07	3.72	3.47	3.34	3.54	2.97	2.15	2.20	2.85	3.34	3.45	3.12		
	HDD10.0	23	8	4	0	0	0	0	0	0	0	0	1	11		
	HDD18.3	518	145	100	35	8	1	0	0	0	0	3	50	176		
	CDD10.0	4886	123	147	278	358	469	634	748	741	620	437	237	94		
	CDD18.3	2339	1	10	55	116	212	384	489	482	370	182	36	1		
	CDH23.3	37802	40	193	853	1736	3231	6543	8462	8234	5704	2306	476	24		
(13)	CDH26.7	24389	2	37	303	818	1862	4512	6043	5828	3664	1172	147	1		
(14)	Wind	WSAvg	2.3	1.9	2.0	2.3	2.8	2.6	2.7	2.4	2.1	1.8	1.7	1.8		
(15) Precipitation	PrecAvg	96	12	15	11	3	1	1	6	10	12	6	4	12		
	PrecMax	201	72	47	51	27	14	5	20	26	78	45	32	53		
	PrecMin	15	0	0	0	0	0	0	0	0	1	0	0	1		
	PrecStd	52	18	15	13	6	3	1	6	8	17	11	8	17		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	26.3	29.9	34.3	37.3	40.6	45.3	45.1	44.6	42.3	37.5	32.3	25.6		
		MCWB	11.1	12.5	14.6	15.1	16.2	19.0	20.5	20.7	19.4	18.0	14.3	12.3		
	2%	DB	24.5	27.8	31.7	35.1	38.9	43.1	43.3	43.3	40.8	35.7	30.2	23.8		
		MCWB	10.5	12.2	13.6	14.6	15.9	18.7	20.6	21.0	19.6	16.8	13.7	11.4		
	5%	DB	22.8	25.8	30.0	33.3	36.9	41.7	42.1	42.1	39.8	34.3	28.5	21.6		
		MCWB	10.1	11.6	13.0	14.2	15.6	18.4	20.5	20.9	19.5	16.1	13.1	10.6		
	10%	DB	21.1	23.9	28.2	31.4	35.0	40.2	41.0	40.8	38.5	32.8	26.2	19.6		
		MCWB	9.5	11.2	12.4	13.4	15.2	17.8	20.5	20.8	19.8	16.0	12.5	9.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.5	15.2	15.2	16.6	17.8	22.1	24.3	24.9	24.5	21.7	16.7	14.9		
		MCDB	19.2	23.9	29.7	32.1	35.6	36.7	33.4	34.8	31.5	27.6	23.4	20.4		
	2%	WB	13.5	13.7	14.4	15.4	16.8	21.3	23.7	24.2	23.7	20.1	15.3	13.8		
		MCDB	17.5	22.9	28.9	31.6	34.9	36.1	34.2	33.9	32.0	29.9	25.1	19.9		
	5%	WB	12.1	12.7	13.7	14.8	16.1	20.3	23.2	23.8	23.0	18.9	14.4	12.7		
		MCDB	17.9	22.2	28.0	30.6	33.9	36.8	34.8	34.2	32.0	29.4	25.1	19.0		
	10%	WB	10.9	11.9	12.9	14.2	15.5	19.1	22.7	23.3	22.4	17.8	13.5	11.5		
		MCDB	18.8	22.0	26.7	29.3	32.8	37.5	35.2	34.5	32.4	28.6	24.4	17.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	13.6	14.0	15.1	15.5	15.6	16.5	13.0	13.1	13.7	14.8	14.3	12.7		
		MCDBR	16.6	16.9	17.7	18.5	18.5	17.4	14.4	14.5	15.4	16.6	16.8	15.3		
		MCWBR	7.8	7.4	7.0	7.2	6.9	6.2	4.5	4.5	5.1	6.8	7.6	7.6		
	5% WB	MCDBR	11.6	14.0	16.2	16.2	16.8	14.7	11.1	11.6	11.2	12.2	14.1	11.9		
		MCWBR	6.0	6.2	6.6	6.5	6.2	5.8	3.7	3.9	4.6	5.6	7.4	6.4		
(40) Clear-Sky Solar Irradiance	taub		0.283	0.287	0.306	0.319	0.330	0.337	0.381	0.378	0.341	0.315	0.291	0.283		
	taud		2.535	2.540	2.456	2.425	2.413	2.404	2.393	2.387	2.463	2.486	2.528	2.548		
	Ebn at Noon		947	976	975	969	954	942	898	899	925	929	928	928		
	Edh at Noon		82	91	107	115	118	119	119	118	105	94	82	77		
(44) All-Sky Solar Radiation	RadAvg		3.55	4.54	5.98	7.33	8.15	8.43	7.57	7.04	6.24	5.06	3.86	3.21		
	RadStd		0.17	0.29	0.20	0.17	0.19	0.29	0.26	0.26	0.20	0.19	0.16	0.16		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (7 neighbors)	N/A	N/A	-1.56	N/A	-0.35	N/A	N/A	N/A	N/A

Nomenclature: See separate page